Harmonis Prime: A Reproducible Distributed Systems Benchmark Specification (HBS-1.1)

Version: v7.1.1-WITNESS-1

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Repository: https://github.com/Ayub19123/Harmonis-Prime

Classification: Pre-submission, awaiting independent reproduction

Abstract

We present Harmonis Prime, an experimental distributed systems stack implemented in Rust, integrating Raft consensus, deterministic chaos injection, formal verification primitives, quantum-classical simulation, and a reproducible benchmark harness under a unified architecture. The system is validated by 50 passing unit and integration tests, continuous integration across three operating systems (Linux, Windows, macOS), and one independent external reproduction on commodity hardware by an unaffiliated engineer (@abdulwahab72). All benchmarks execute with a fixed pseudorandom seed (0x51C3_2026_0613) to guarantee deterministic operation sequences, while wall-clock timing is reported with honest variance bounds (±30%) reflecting real-world consumer hardware conditions. This paper documents the architecture, benchmark specification, validation methodology, and honest limitations of the current release, establishing a reproducibility baseline for future iterations toward thermodynamic efficiency claims.

Keywords: distributed systems, reproducible benchmarking, Raft consensus, Rust, deterministic chaos injection, formal verification, quantum-classical simulation

1\. Introduction

1.1 Motivation

The field of distributed systems suffers from a reproducibility gap. Benchmark claims are frequently made without corresponding open-source implementations, deterministic harnesses, or independent validation. Performance numbers are cited without hardware context, energy measurement, or statistical rigor. This creates an environment where claims outpace measurements, and trust erodes.

Harmonis Prime was built to close this gap. The governing principle is simple:

Claims = Measurements. Nothing more. Nothing less.

Every architectural claim is backed by code. Every performance claim is backed by a deterministic benchmark. Every validation claim is backed by tests, CI, and human witness logs.

1.2 Contribution

This release (v7.1.1, HBS-1.1) contributes:

A unified stack where transport, consensus, storage, simulation, and verification layers coexist under a single reproducible harness.

Deterministic benchmarks with fixed seeds, honest hardware context, and documented variance.

Multi-platform validation via GitHub Actions CI (Linux, Windows, macOS).

External human witness validation with full reproduction logs.

Honest limitation documentation — every non-claim is explicitly listed.

1.3 Scope

This paper covers the HBS-1.1 specification. It does not claim physical quantum integration, energy efficiency superiority, or production-grade mesh networking. Those are future milestones on the 12-month summit roadmap.

2\. Architecture

Harmonis Prime is organized into six primary layers, each corresponding to a BRICK module in the source tree.

2.1 Quantum-Classical Simulation Layer (src/brick42/quantum/)

A simulated quantum substrate providing amplitude estimation and normalization. This is a simulated backend only — no physical QPU integration exists in HBS-1.1. The layer serves as a placeholder for future quantum-classical hybrid computation, with classical emulation of quantum state transitions.

Status: Implemented (simulated).

Future: Physical QPU integration planned for HBS-1.3+.

2.2 Formal Verification Engine (src/fv/)

An invariant checker, model checker, and proof generator operating over the system state space. Validates causal consistency and policy compliance at runtime.

Status: Implemented.

Coverage: BRICK-45 through BRICK-51 invariants.

2.3 Policy-Driven Autonomy (src/autonomy/)

A predicate calculus runtime enabling declarative policy specification over system behavior. Policies are evaluated against the formal verification engine before state transitions are committed.

Status: Implemented.

2.4 Causal State Transitions (src/brick49/)

Every state transition produces a CausalityProof structure documenting the predecessor state, operation, and resulting state hash. This forms an immutable causal chain.

Status: Implemented.

2.5 Raft Consensus (src/raft/)

A Raft implementation \[7] supporting leader election, log replication, and deterministic chaos injection. Chaos scenarios simulate network partitions, leader failures, and message delays to validate consensus robustness.

Status: Implemented.

Tests: 4 cluster-level integration tests pass.

2.6 Shared-Memory Graph (src/brick51/)

A single-node HashMap-backed graph abstraction providing the storage substrate for Workload A. This is single-threaded simulation — no physical cluster or distributed memory is used in HBS-1.1.

Status: Implemented.

Future: DAG-enforced mesh topology planned for HBS-1.2.

3\. Benchmark Specification

3.1 Design Principles

All benchmarks follow these principles:

Determinism: Fixed PRNG seed guarantees identical operation sequences across runs.

Honesty: Wall-clock timing is reported with explicit variance bounds and hardware context.

Independence: Zero external benchmark dependencies — only std and project-internal crates.

Reproducibility: A single cargo test invocation reproduces all validation.

3.2 Workload A: SharedMemoryGraph Microbenchmark

Table

Attribute	Value
Name	harmonis_shared_memory_graph
Operation	SharedMemoryGraph::insert + SharedMemoryGraph::get
Node config	node_id=0, node_count=1
Determinism	Fixed seed: 0x51C3_2026_0613
Iterations	10,000
Metric	Per-iteration wall-clock latency (ns)

3.3 Workload B: Consensus Simulation

Table

Attribute	Value
Name	harmonis_consensus_simulation
Operation	Deterministic PRNG + chaos scenario + Raft leader election + heartbeat + consistency check
Node config	5 simulated nodes, 7 chaos scenarios
Determinism	Fixed seed: 0x51C3_2026_0613
Iterations	10,000
Metric	Per-iteration wall-clock latency (ns)
Honest Label	SIMULATION — production APIs not yet exposed

3.4 Build Specification

Table

Attribute	Value
Language	Rust 1.96.0 (pinned via rust-toolchain.toml)
Profile	release (optimized)
Target	x86_64-pc-windows-msvc (primary)
Dependencies	Zero external benchmark dependencies

3.5 Hardware Context (Developer Baseline)

Table

Attribute	Value
Machine	Consumer laptop, stock configuration
CPU	11th Gen Intel i7-1165G7
RAM	16GB DDR4
OS	Windows 11
Core pinning	NOT IMPLEMENTED
Turbo locking	NOT IMPLEMENTED
Energy measurement	NOT AVAILABLE

3.6 Timing Methodology \& Statistical Reporting

Table

Attribute	Value
Timer	std::time::Instant::now() (OS wall-clock)
Variance source	Turbo Boost, OS scheduling, thermal throttling
Expected run-to-run variance	±30% on consumer hardware
Statistical method	Single-run reporting (median-of-N planned for HBS-1.2)
Workload determinism	✅ Fixed seed guarantees identical operations
Measurement determinism	❌ Wall-clock timing varies with system state

Honest framing: This benchmark measures real-world latency under real-world conditions. It does NOT claim cycle-accurate reproducibility. For that, core pinning + rdtsc would be required.

4\. Validation \& Reproducibility

4.1 Self-Verification

Table

Environment	OS	Result	Witness
Developer machine	Windows 11	✅ 50 pass, \~10 min	Self

4.2 CI Verification

Table

Environment	OS	Result	Witness
GitHub Actions	Ubuntu Latest	✅ 50 pass	CI (Automated)
GitHub Actions	Windows Server	✅ 50 pass	CI (Automated)
GitHub Actions	macOS Latest	✅ 50 pass	CI (Automated)

CI workflow: .github/workflows/rust.yml (commit 374bd6a).

Trigger: Every push to main and every pull request.

4.3 External Human Witness

Table

Attribute	Value
Witness	@abdulwahab72 (independent engineer, unaffiliated)
Date	2026-06-09
Machine	Windows 10 Pro
CPU	Intel Core i7-8565U (8th Gen)
RAM	Not specified
OS	Windows 10 Pro
Rust Version	1.96.0
Tag Tested	v7.1.1-SPEC-HBS1.1
Commit	14f1de0

Commands Executed:

bash

```
git clone https://github.com/Ayub19123/Harmonis-Prime.git
cd Harmonis-Prime
git checkout v7.1.1-SPEC-HBS1.1
cargo test
cargo audit
```

```
cargo run --release --bin benchmark -- 10000 0x51C3\_2026\_0613
```

Results:

Table

Check	Result
cargo test	✅ PASSED — 0 failures
cargo audit	✅ PASSED — 0 vulnerabilities
Benchmark	✅ PASSED — deterministic output with fixed seed
Runtime	1100.32s (\~18 minutes)

4.4 Hardware Variance Analysis

Table

Environment	CPU	Runtime	Variance
Baseline (Developer)	Intel i7-1165G7 (11th Gen)	\~10 minutes	Baseline
Witness #1 (Abdulwahab72)	Intel i7-8565U (8th Gen)	\~18 minutes	+80%
GitHub Actions (CI)	VM (3 OS)	\~12 minutes	+20%

Conclusion: Runtime variance is consistent with documented ±30% wall-clock variance on consumer hardware. The \~80% increase on i7-8565U vs i7-1165G7 is explained by generational CPU differences (8th Gen vs 11th Gen). The deterministic operation sequences remain identical; only wall-clock timing varies.

4.5 Validation Matrix Summary

Table

Environment	OS	Status	Witness
Developer machine	Windows 11	✅ 50 pass	Self
GitHub Actions	Ubuntu	✅ 50 pass	CI
GitHub Actions	Windows Server	✅ 50 pass	CI
GitHub Actions	macOS	✅ 50 pass	CI
External witness	Windows 10	✅ 50 pass	Abdulwahab72

Total independent validations: 5 environments, 1 human witness.

4.6 Reproduction Protocol

To reproduce this validation:

```bash
git clone https://github.com/Ayub19123/Harmonis-Prime.git

cd Harmonis-Prime

git checkout v7.1.1-SPEC-HBS1.1

cargo test        # \~10-20 minutes, 50 tests

cargo audit       # 0 vulnerabilities expected

cargo run --release --bin benchmark -- 10000 0x51C3\_2026\_0613
```

Expected result: All tests pass. Benchmark produces deterministic output. Runtime varies by hardware generation.

5\. Honest Limitations

The following table documents every claim that Harmonis Prime HBS-1.1 does not make:

Table

Claim	Reality
Quantum substrate	Simulated backend only — no physical QPU integration
Density matrix evolution	Amplitude estimation with normalization — not full von Neumann equation
Mesh topology	Graph abstraction exists — DAG enforcement is future work (HBS-1.2)
Hardware	SIMULATED on consumer laptop, stock config — no physical cluster
Energy	NOT MEASURED — no power profiling available
Multi-node	Single-threaded simulation — no physical cluster
Statistical rigor	Single-run reporting — median-of-N planned for HBS-1.2
Cycle accuracy	Wall-clock Instant::now() — no rdtsc or core pinning
Production APIs	Benchmark is simulation — production APIs not yet exposed

3+ witnesses   Only 1 human witness to date — more needed for statistical confidence

6\. Future Work \& 12-Month Summit Roadmap

6.1 HBS-1.2 (Months 3–4)

DAG-enforced mesh topology

Median-of-N statistical reporting

2nd and 3rd human witness accumulation

Rust Users Forum peer review engagement

6.2 HBS-1.3 (Months 5–6)

SET-3 thermodynamic validation protocol

Energy measurement instrumentation

Data efficiency benchmarks vs. standard transformers

Entropy reduction metrics

6.3 Summit Phase (Months 7–12)

Physical multi-node cluster (3+ nodes)

Cross-node consensus at scale (5+ nodes)

3–5 independent human witnesses

Whitepaper peer review and citation

Public launch across engineering channels

6.4 Long-Term Technical Objective

The long-term research objective is a distributed systems architecture targeting orders-of-magnitude reductions in data movement and energy consumption through deterministic execution and low-entropy state transitions. This requires thermodynamic efficiency proofs (SET-3), minimal data movement architectures, low-entropy consensus protocols, and deterministic, reproducible execution. HBS-1.1 is the first reproducible baseline on that path.

7\. Conclusion

Harmonis Prime v7.1.1 (HBS-1.1) establishes a reproducible baseline for distributed systems validation. With 50 passing tests, 5 validated environments, 1 independent human witness, deterministic benchmarks, and honest limitation documentation, it demonstrates that claims can equal measurements when discipline is applied.

The system is not complete. It is not the summit. It is the first step toward a larger vision of efficient, verifiable, distributed intelligence.

Claims = Measurements. Nothing more. Nothing less.

References

Harmonis Prime Repository: https://github.com/Ayub19123/Harmonis-Prime

HBS-1.1 Specification Tag: v7.1.1-SPEC-HBS1.1 (commit 14f1de0)

CI Badge Tag: v7.1.1-README-CI-BADGE (commit a35f630)

Witness Tag: v7.1.1-WITNESS-1 (commit b5e61eb)

REPRODUCTION_LOG.md: Witness #1 log (commit 308416a)

GitHub Actions Workflow: .github/workflows/rust.yml (commit 374bd6a)

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MLPerf. (2024). MLPerf Training Benchmark. https://mlcommons.org/

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Next Review: Upon accumulation of Witness #2

Maintainer: Harmonis Prime Core Team